2

1. Han--Li--Sun $\beta = 0$ Wang Proposition 5.2
“ 2”

2. $\beta > 0$ Han--Li--Sun White
$\Theta < 2$
Ψ $\Psi < 1 + \varepsilon_{\mathrm{HLS}}$

1.

M^4 Kähler $M \hookrightarrow \mathbb{R}^N$

$$F : \Sigma \times [0, T) \longrightarrow M$$

$T < \infty$

$$\eta = *F_t^* \omega = \cos \alpha.$$

(X_0, T) ψ $\psi \equiv 1$ X_0

$$\rho_{X_0,T}(Y, t) = \frac{1}{4\pi(T-t)} \exp\left(-\frac{|Y - X_0|^2}{4(T-t)} \right).$$

$$\Theta(X_0, T) := \lim_{t \uparrow T} \int_{\Sigma_t} \psi \, \rho_{X_0,T} \, d\mu_t. \tag{1.1}$$

Huisken $(0,0)$
X_0

Σ_{I} (X_0, T) $x_j \in \Sigma$ $t_j \uparrow T$

$$F(x_j, t_j) \to X_0, \qquad |A|(x_j, t_j) \to \infty,$$

T

$$\sup_{\Sigma_t} |A|^2 \leq \frac{C}{T-t}, \tag{1.2}$$

Σ_{I}

2.

2.1 2

M^4 Kähler $F : \Sigma \times [0, T) \to M$ T

1. $\delta > 0$ $\eta = *F_t^* \omega \geq \delta$ $\Sigma \times [0, T)$
2. $C < \infty$ (1.2)
3. $(X_0, T) \in \Sigma_{\mathrm{I}}$

$$\Theta(X_0, T) < 2. \tag{2.1}$$

$\varepsilon > 0$ F $\Sigma \times [0, T + \varepsilon)$

T

$$\limsup_{t \uparrow T} \sup_{\Sigma_t} |A| = \infty. \tag{2.2}$$

$x_j \in \Sigma$ $t_j \uparrow T$ M

$$F(x_j, t_j) \longrightarrow X_0 \in M.$$

(1.2) $(X_0, T) \in \Sigma_{\mathrm{I}}$

(X_0, T)

$$F_s^{\lambda}(x) = \lambda\big(F(x, T + \lambda^{-2}s) - X_0\big), \qquad s < 0, \tag{2.3}$$

Brakke $\mathcal{T}$ Wang Proposition 5.2

Kähler

$$V = \sum_{j=1}^{J} m_j |P_j|, \qquad m_j \in \mathbb{N}, \tag{2.4}$$

$P_j \subset T_{X_0} M \simeq \mathbb{R}^4$ “

” “ ” $\eta \geq \delta$ Wang

$\Theta < 2$

1

$$\frac{1}{4\pi}\int_{P_j} e^{-|Y|^2/4}\, d\mathcal{H}^2(Y) = \frac{1}{4\pi}\int_{\mathbb{R}^2} e^{-|y|^2/4}\, dy = 1. \tag{2.5}$$

(2.4)

$$\Theta(X_0, T) = \frac{1}{4\pi}\int e^{-|Y|^2/4}\, d\|V\|(Y) = \sum_{j=1}^{J} m_j. \tag{2.6}$$

(2.1) 1 $J = 1$ $m_1 = 1$

$$\Theta(X_0, T) = 1. \tag{2.7}$$

White

White $\varepsilon_W > 0$ $1 + \varepsilon_W$
(2.7)

$$\Theta(X_0, T) = 1 < 1 + \varepsilon_W,$$

(X_0, T) (2.2) (X_0, T)

T $|A|$ $[0, T)$
$\varepsilon > 0$ $[0, T + \varepsilon)$

2.2

$\Theta \le 2$ P $2|P|$ 2
“ ”

2.3 Wang Proposition 5.2

Wang multiplicity one plane White “ ”
$\Theta < 2$ (2.6)
“ ” 2

3. Han--Li--Sun β-

Han--Li--Sun

$$\partial_t F = f = \frac{\cos^2\alpha\, H - \beta\sin^2\alpha\, V}{\cos^2\alpha + \beta\sin^2\alpha}, \qquad \beta \ge 0, \tag{3.1}$$

$V = \partial_2\alpha\, v_3 + \partial_1\alpha\, v_4$ $\beta = 0$ (3.1) $\partial_t F = H$

$$\Psi(X_0,t_0;t)=\int_{\Sigma_t}\frac{1}{\cos^p\alpha}\,\phi(F),\rho_{X_0,t_0}(F,t)\,d\mu_t,\qquad p\geq p_0, \tag{3.2}$$

- Corollary 4.2 $\varepsilon_{\text{HLS}}>0$ $r>0$
$$\Psi(X_0,t_0;t_0-r^2)<1+\varepsilon_{\text{HLS}}, \tag{3.3}$$
(X_0,t_0) (3.1)
- Theorem 3.1 $|A|$ (3.1)

3.1 β-

F (3.1) T $\cos\alpha\geq\delta>0$ X_0 $r_{X_0}>0$ (3.3) $(t_0,r)=(T,r_{X_0})$ T

Corollary 4.2 (X_0,T) $|A|$ $t\uparrow T$ M X_0 $|A|$ Theorem 3.1

4. $\beta>0$ (3.3) $\Theta<2$

1. White $\beta>0$ (3.1) H
2. Han--Li--Sun Theorem 5.7 λ- Kähler
3. Kähler $c=\cos\alpha_P\in(0,1]$ 1 (3.2) c^{-p}
$$\Theta<2$$
varifold
$$\Psi<1+\varepsilon_{\text{HLS}}.$$
4. Wang Proposition 5.2 $\beta>0$
$\Theta<2$

- $\cos\alpha_P = 1$
- (3.1) “ ”
- (3.3) $1 + \varepsilon_{\mathrm{HLS}}$

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5.

$\beta = 0$ MCF	White	$\Theta < 2$ Wang	Θ 2 1
$\beta > 0$ Han--Li--Sun	Han--Li--Sun Corollary 4.2	$\Psi < 1 + \varepsilon_{\mathrm{HLS}}$	Theorem 3.1
$\beta > 0$ $\Theta < 2$	White	1	

6.

2.1

> $F_t : \Sigma \to M^4$ Kähler T $*F_t^*\omega \geq \delta > 0$ $|A_t|^2 \leq C/(T-t)$ (X_0, T) $\Theta(X_0, T) < 2$ $\varepsilon > 0$ $[0, T+\varepsilon)$

$\beta > 0$ 3.1 $\Psi < 1 + \varepsilon_{\mathrm{HLS}}$ 4 $\Theta < 2$

[1] M.-T. Wang, *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds*, J. Differential Geom. 57 (2001), 301--338 Proposition 5.2

[2] X. Han, J. Li and J. Sun, *Gradient Flow for β-Symplectic Critical Surfaces*, Ann. Inst. H. Poincaré C Anal. Non Linéaire 41 (2024), 1083--1116 Theorems 3.1, 4.1, 5.7 Corollary 4.2

[3] B. White, *A Local Regularity Theorem for Mean Curvature Flow*, Ann. of Math. 161 (2005), 1487--1519